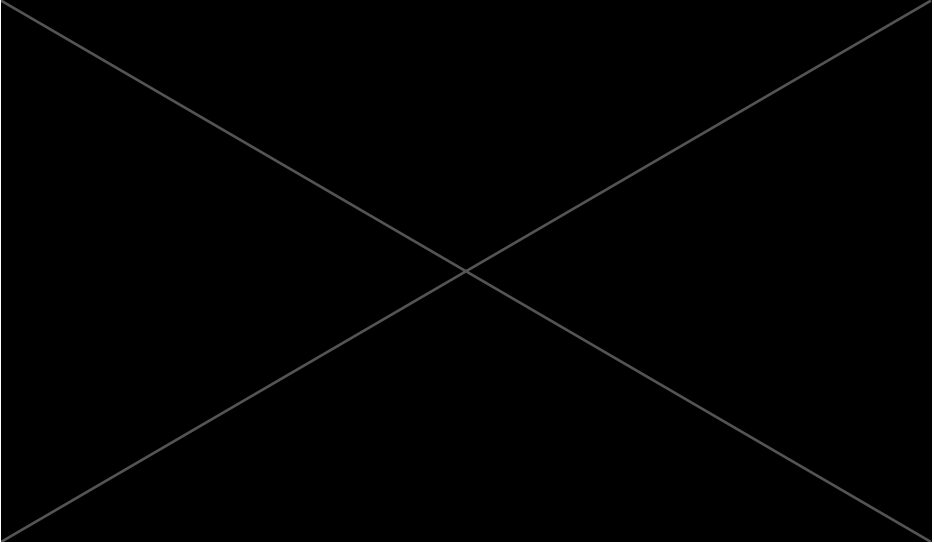




11.9 Optimal Event Trading

Consider a single asset with price $P(t)$. We assume that the asset is infinitely divisible (we can buy arbitrarily small amounts of it) and that it can be shorted at zero cost in arbitrary amounts. The investor holds $x(t)$ dollars' worth of the asset. The trading volume is varying over time. We assume it is constant: "time" in this section refers to "volume time": the total traded volume in a time interval is proportional to the interval length.³ Therefore, physical time intervals in which volume is high correspond to longer intervals in volume time. The return process in the interval $[0, T]$ is

$$dP(t) = P(t)[\alpha(t)dt + \sigma dB(t)]$$

$$P(0) = 1$$

³ Volume time can be computed based on realized or forecasted trading volume $v(t)$. Let $V(t) = \int_0^t v(s)ds$, and define $\tau = V(t)$. The volume traded in the interval $[\tau, \tau + \delta]$ is constant. It is equal to the volume traded in the interval $[V^{-1}(\tau), V^{-1}(\tau) + (V^{-1})'(\tau)\delta]$, which is

$$V'(V^{-1}(\tau)) \times [(V^{-1})'(\tau)\delta] = V'(V^{-1}(\tau)) \times \left[\frac{\delta}{V'(V^{-1}(\tau))} \right] = \delta$$

We make two assumptions:

1. *Alpha curve*: The function $\alpha(t)$ is known in $[0, T]$.
2. *Quadratic transaction costs*: The transaction costs associated to trading a dollar amount v of the stock are proportional to v^2 . We write

$$TC(v) = \frac{\kappa^2}{2} v^2 \quad (11.8)$$

The general problem is

$$\begin{aligned} \max_x \quad & \int_0^T \left(\alpha(t)x(t) - \frac{\kappa^2}{2} \dot{x}^2(t) - \frac{\rho\sigma^2}{2} x^2(t) \right) dt \\ \text{s.t.} \quad & x(0) = 0 \\ & x(T) = 0 \end{aligned}$$

where

- αx is the expected return;
- $(\kappa^2/2)\dot{x}^2$ is the transaction cost;
- $(\rho\sigma^2/2)x^2$ is the risk aversion term.

Before producing examples and analytical processes, we discuss the model assumptions and their limits. We assume that the function α is known by the investor. This seems a reasonable assumption, since the investor does have a view of the stock returns over the horizon. However, we do not model two important components of the investor's information. First, the investor may have different conviction for the stock alpha at a given time. If the stock has an earning announcement, then the α view on that date will be stronger than for preceding days. The second assumption we make is that no new information is acquired during the investment and α is not updated based on the information. However, new information arrives all the time, both in the form of company-specific news and market participant information. The stock may experience a run-up, and this run-up may be interpreted as other investors acting on the same information contained in α . In this event, we would either need to update α , or model the investment problem in an altogether different way. The assumption of quadratic transaction costs is also an idealization. Transaction

cost models usually include a linear term and a polynomial one, which has an exponent between $2/3$ and 2 . The assumption is not unrealistic because the exponent estimated from empirical data is closer to 2 , and the linear term is much smaller than the polynomial one. We also make the assumption that κ is constant. This seems acceptable, so long as market conditions do not change; typical horizons for this problem are a month or less.

Optimal Policy. Write the objective function as

$$\int_0^T L(t, x, \dot{x}) dt, \text{ where } L(t, x, \dot{x}) = \alpha x - \frac{\kappa^2}{2} \dot{x}^2 - \frac{\rho^2 \sigma^2}{2} x^2$$

The Euler-Lagrange equation is

$$\partial_x L - \frac{d}{dt} \partial_{\dot{x}} L = 0, \text{ i.e., } \kappa^2 \ddot{x} = -\alpha + \rho^2 \sigma^2 x$$

This is a non-homogeneous linear ordinary differential equation. Let us define

$$a = \alpha / \kappa^2$$

$$b = \rho \sigma / \kappa$$

$$F(t) = \int_0^t \frac{a(\zeta) e^{b\zeta}}{2b} d\zeta$$

$$G(t) = \int_0^t \frac{a(\zeta) e^{-b\zeta}}{2b} d\zeta$$

The solution is given by

$$x(t) = e^{-bt}(F(t) + c) - e^{bt}(G(t) + c)$$

$$c = \frac{e^{-bT}F(T) - e^{bT}G(T)}{e^{bT} - e^{-bT}}$$

The trading rate $\dot{x}$ is more instructive, and is equal to

$$\dot{x}(t) = e^{-bt}[\dot{F}(t) - bF(t) - bc] - e^{bt}[\dot{G}(t) + bG(t) + bc]$$

The formulas hold for any forecasted time-varying alpha. We are interested in the special case of *event-based return*:

$$\alpha(t) = \alpha_0 \delta(t - t_0)$$

where $\delta(\cdot)$ is Dirac's delta.⁴ There is a single event, e.g., an earning announcement. In this case

$$\begin{aligned}
 F(t) &= \frac{\alpha_0 e^{bt_0}}{2\sigma\rho\kappa} \theta(t - t_0) = K_F \theta(t - t_0) \\
 G(t) &= \frac{\alpha_0 e^{-bt_0}}{2\sigma\rho\kappa} \theta(t - t_0) = K_G \theta(t - t_0) \\
 c &= \frac{e^{-bT} K_F - e^{bT} K_G}{e^{bT} - e^{-bT}} \\
 \dot{x}(t) &= e^{-bt} \left(\frac{\alpha_0 e^{bt_0}}{2\sigma\rho\kappa} \delta(t - t_0) - bF - bc \right) - \\
 &\quad e^{bt} \left(\alpha_0 \frac{e^{-bt_0}}{2\sigma\rho\kappa} \delta(t - t_0) + bG + bc \right) \\
 &= -be^{-bt}(F + c) - be^{bt}(G + c) \\
 &= \begin{cases} -bc(e^{-bt} + e^{bt}) & \text{if } t \leq t_0 \\ -be^{-bt}(K_F + c) - be^{bt}(K_G + c) & \text{if } t > t_0 \end{cases}
 \end{aligned}$$

Note that $K_G = e^{-2bt_0} K_F$ and $c < 0$. As an example, assume that $\alpha_0 > 0$. The trading policy is to trade on the side of α_0 up to the event. The trade is not linear. After t_0 , trading is in the opposite direction:

$$\begin{aligned}
 K_G &= e^{-2bt_0} K_F \\
 K_F + c &\geq K_G + c \\
 &= K_G(e^{bT} - e^{-bT}) - (e^{-bT} K_F - e^{bT} K_G) \\
 &= K_G e^{bT} [1 - e^{-2b(T-t_0)}] \\
 &\geq 0
 \end{aligned}$$

so that both terms in the formula for $t \geq t_0$ are negative.

Since there is no natural liquidation horizon T , it is useful to consider for practical applications the case $T \rightarrow \infty$.

⁴ Dirac's delta $\delta(\cdot)$ is a generalized function equal to 0 for $x \neq 0$ and such that $\int_{-\infty}^{\infty} \delta(x) dx = 1$. An associated function is Heavyside's function $\theta(\cdot)$, which is equal to 0 for $x < 0$ and to 1 for $x \geq 0$.

Optimal Policy with Long-Horizon Liquidation. In this limit we have $c \rightarrow -K_G$, and

$$\begin{aligned} x(t) &= e^{-bt}(F(t) - K_G) - e^{bt}(G(t) - K_G) \\ &= \begin{cases} \frac{\alpha_0}{b\kappa^2} e^{-bt_0} \sinh(bt) & \text{if } t \leq t_0 \\ \frac{\alpha_0}{b\kappa^2} \sinh(bt_0) e^{-bt} & \text{if } t > t_0 \end{cases} \end{aligned}$$

The optimal policy is to build the position and then to liquidate it, at a fast rate right after the event, and then more and more slowly. The optimal position at event date t_0 is

$$x(t_0) = \frac{\alpha_0}{2b\kappa^2} (1 - e^{-2bt_0})$$

The optimal trading rate is

$$\dot{x}(t) = \begin{cases} \frac{\alpha_0 e^{-bt_0}}{\kappa^2} \cosh(bt) & \text{if } t \leq t_0 \\ -\frac{\alpha_0}{\kappa^2} \sinh(bt_0) e^{-bt} & \text{if } t > t_0 \end{cases}$$

Just prior to the event and after the event the trading rate is identical in size but opposite in direction:

$$\text{trading rate just before } t = t_0 : \frac{\alpha_0}{2\kappa^2} (1 + e^{-2bt_0})$$

$$\text{trading rate just after } t = t_0 : -\frac{\alpha_0}{2\kappa^2} (1 + e^{-2bt_0})$$

Optimal Policy with Long-Horizon Liquidation and Risk-Neutral Investor. In the limit $\rho \rightarrow 0$, we have

$$x(t) \simeq \begin{cases} \frac{\alpha_0}{\kappa^2} t & \text{if } t \leq t_0 \\ \frac{\alpha_0 t_0}{\kappa^2} (1 - t) & \text{if } t > t_0 \end{cases} \quad (11.9)$$